Fighting 'Ozempic Butt': Can a New Drug Preserve Muscle During Weight Loss?

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The rise of **GLP-1 receptor agonists** like Ozempic, Wegovy, and Mounjaro has been a game-changer for obesity treatment, but they come with an unexpected downside: significant muscle loss. Research suggests up to a third of the weight shed on these drugs can be **lean mass**, not just fat. For some users, this leads to a deflated, saggy appearancecolloquially dubbed "Ozempic butt"and a spike in consultations with plastic surgeons. Now, a study published in Nature Medicine offers a potential solution. The drug **apitegromab**, given as an **intravenous infusion**, blocks **myostatin**, a protein that inhibits muscle growth. In a trial of 102 adults, mostly women, those who received **apitegromab** alongside their obesity medication preserved more muscle while still losing fat, according to body scans. The company behind the drug, which funded the research, is exploring a self-injection pen to make it as convenient as the weight-loss jabs themselves. But experts urge caution. Dr. Marie Spreckley of Cambridge University, who was not involved in the study, calls the findings "encouraging early evidence" rather than definitive proof. She and others emphasize the need for larger, longer trials to see if preserving muscle actually improves strength, well-being, and long-term healthnot just body shape. Dr. Brendan Gabriel from the University of Aberdeen adds that the treatment might not be for everyone, but could benefit those losing muscle mass unusually fast. Currently, **apitegromab** is only available in clinical trials and is also being studied for muscle-wasting diseases like **spinal muscular atrophy**. For now, the standard advice remains: people on GLP-1s should eat a balanced diet and engage in strength traininganything that makes muscles work harder, from gym sessions to carrying groceries. And health experts strongly warn against using these powerful drugs for cosmetic quick fixes. The quest for a healthier weight, it seems, still requires more than a magic injection.

Word Treasure

GLP-1 receptor agonists -- GLP-1 receptor agonists are a class of medicines (like Ozempic and Wegovy) that help control blood sugar and appetite, often leading to weight loss.

myostatin -- Myostatin is a natural protein in the body that puts a brake on muscle growth. Blocking it can allow muscles to get bigger.

apitegromab -- Apitegromab is an experimental drug that blocks myostatin, and researchers are testing if it can preserve muscle during weight loss.

lean mass -- Lean mass is the weight of everything in the body that isn't fat -- mainly muscles, bones, and organs.

intravenous infusion -- An intravenous infusion means giving medicine directly into a vein through a tube, so it enters the bloodstream quickly.

spinal muscular atrophy -- Spinal muscular atrophy is a genetic disease that makes muscles very weak because nerve cells in the spine die, and apitegromab is also being studied to help these patients.

Background you need

Ozempic (whose active ingredient is semaglutide) was originally approved in 2017 for type 2 diabetes, but its dramatic weight-loss effects quickly turned it into a global phenomenon -- so much so that shortages occurred for diabetes patients. The term 'Ozempic butt' appeared on social media as users noticed sagging skin and muscle loss, which prompted many to seek plastic surgery. Behind the scenes, scientists have studied myostatin inhibitors for over two decades; natural myostatin mutations in cattle and dogs result in extreme muscle bulk, and earlier drug trials for muscle-wasting conditions were sometimes disappointing. Now, with the obesity drug boom, researchers see a new opportunity. The apitegromab trial was funded by the drug's maker (Scholar Rock) and published in the prestigious journal Nature Medicine, but the small size and short duration mean it is only an early step.

Break it down

LEAD: GLP-1 drugs revolutionized obesity treatment but come with an unexpected downside -- substantial muscle loss, nicknamed 'Ozempic butt'.

+- statistic: up to one-third of lost weight can be lean mass.

HOPE FROM A NEW STUDY: Apitegromab, a myostatin blocker, shows promise in preserving muscle during weight loss.

+- METHOD: 102 adults, mostly women, received the drug as an intravenous infusion alongside their obesity medication.

+- RESULT: body scans showed more muscle preserved while still losing fat.

+- FUTURE FORM: The company is exploring a self-injection pen for convenience.

EXPERT SKEPTICISM: Dr. Marie Spreckley calls it 'encouraging early evidence' but stresses the need for larger, longer trials to prove real benefits to strength and health.

+- DR. BRENDAN GABRIEL: The treatment might help people losing muscle unusually fast.

CURRENT STATUS: Apitegromab is only available in clinical trials and is also being tested for spinal muscular atrophy.

PRACTICAL ADVICE: Experts still recommend a balanced diet and strength training -- not quick fixes.

SOCIETAL WARNING: Health professionals caution against using powerful drugs for cosmetic reasons alone.

Why it matters

The quest for a healthier weight often ignores body composition -- and losing muscle can slow metabolism and harm long-term health. For teenagers, this story highlights why understanding science is crucial before trusting a 'miracle' drug, and that true fitness involves more than numbers on a scale.

Test what you remember

1. **What is a major side effect of GLP-1 drugs like Ozempic mentioned in the article?**
   - Ⓐ Hair loss
   - Ⓑ Significant muscle loss
   - Ⓒ Increased appetite
   - Ⓓ Vision problems

2. **What does the drug apitegromab do?**
   - Ⓐ It speeds up fat burning
   - Ⓑ It blocks myostatin to help muscle grow
   - Ⓒ It reduces hunger signals in the brain
   - Ⓓ It replaces lost muscle with fat tissue

3. **How was apitegromab given in the trial?**
   - Ⓐ As a daily pill
   - Ⓑ Through an intravenous infusion
   - Ⓒ With a self-injection pen
   - Ⓓ As a nasal spray

4. **What was one key result of the apitegromab trial according to the article?**
   - Ⓐ Participants lost more fat than those on placebo
   - Ⓑ Participants preserved more muscle while still losing fat
   - Ⓒ Participants experienced no weight loss at all
   - Ⓓ Participants gained weight instead of losing it

5. **Why do experts like Dr. Marie Spreckley urge caution about the findings?**
   - Ⓐ Because the drug is already widely available and misused
   - Ⓑ Because the trial only had men and no women
   - Ⓒ Because it's early evidence and larger, longer trials are needed
   - Ⓓ Because the drug caused serious heart problems in most people

6. **What is the current recommended way for people on GLP-1 drugs to protect their muscles?**
   - Ⓐ Take apitegromab daily
   - Ⓑ Eat a balanced diet and do strength training
   - Ⓒ Avoid all physical activity
   - Ⓓ Reduce protein intake

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: C 6: B

Questions to think about

1. Positive / hopeful: For patients who need GLP-1 drugs to manage severe obesity but fear becoming frail, a muscle-sparing treatment could make weight loss safer and more sustainable, potentially reducing the need for plastic surgery and improving quality of life.

2. Negative / critical: Critics warn that adding a second injection to an already expensive drug regimen could turn weight loss into an endless medical procedure. Moreover, the long-term side effects of myostatin blockers are unknown, and there's a risk they'll be misused for body shaping by people without medical need.

3. Neutral / analytical: The development of apitegromab reflects a broader pattern in medicine: a successful drug creates a new side effect, and industry races to solve it with another drug. Whether this approach leads to genuine health improvements or simply medicalizes normal aging and body changes remains an open question.

4. Forward-looking: If larger trials confirm apitegromab's effectiveness and safety, regulators may approve the first drug specifically designed to prevent muscle loss during pharmacologic weight loss. That could reshape obesity treatment guidelines and redefine what successful weight management means -- but access and affordability will be major hurdles.

MY THOUGHTS (at least 20 words):